

Industrial Internship Report on “Student Performance Analytics Dashboard”

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was the design and development of a Student Performance Analytics Dashboard — a Flask-based web application that lets a user upload student academic records and instantly view automatically generated performance statistics, built using Python, Flask, SQLite, NumPy and Pandas.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

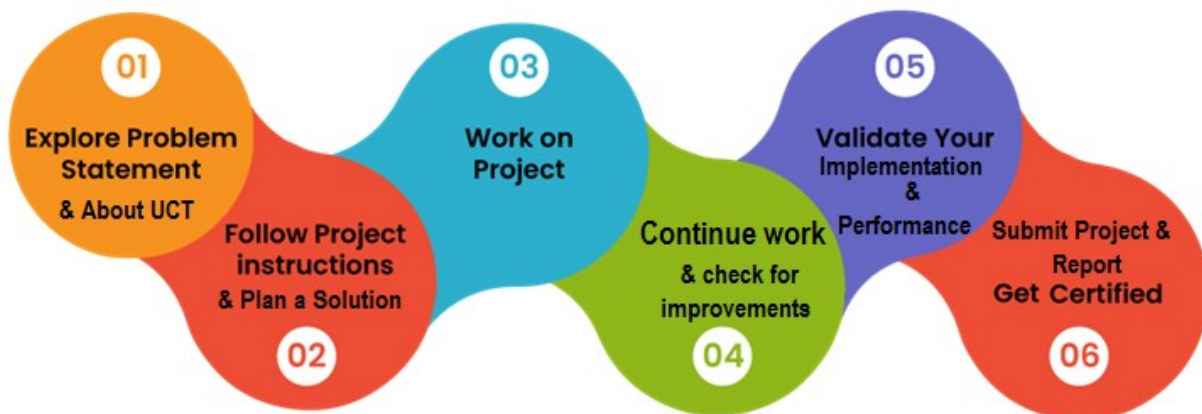
Over the course of this internship, I progressed from Python fundamentals to building a complete Flask-based web application, then strengthened my data-handling skills with NumPy and Pandas, validated my understanding through a proctored assessment, and applied everything to my final project, the Student Performance Analytics Dashboard.

Internships like this one bridge the gap between classroom learning and real-world software development, giving students hands-on exposure to the tools, workflows and problem-solving approaches expected in the industry — exposure that is difficult to gain from coursework alone.

My project, the Student Performance Analytics Dashboard, is a Flask web application that lets a user upload a dataset of student records and instantly view processed statistics — averages, top performers, grade distribution — generated using Pandas and NumPy, addressing the common need for a lightweight tool to analyze academic data without spreadsheet software.

upSkill Campus and UniConverge Technologies Pvt Ltd gave me the opportunity to work on a guided, real project with structured weekly learning resources, assessments and mentorship that helped me build both my technical skills and my confidence in independent problem-solving.

How Program was planned



This internship strengthened my Python fundamentals, taught me how to design and build a Flask application from scratch, and gave me practical experience with NumPy and Pandas for data analysis — overall, it was a rewarding experience that significantly improved both my technical and problem-solving skills.

I would like to thank my mentors and the entire upSkill Campus and UniConverge Technologies (UCT) team for their guidance, structured learning resources and continuous support throughout this internship.

To my juniors and peers: be consistent with practice, don't skip the fundamentals even when you're eager to build projects, and treat every small assignment or quiz as a chance to find and fix gaps in your understanding early.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT)**, **Cyber Security**, **Cloud computing (AWS, Azure)**, **Machine Learning**, **Communication Technologies (4G/5G/LRaWAN)**, **Java Full Stack**, **Python**, **Front end** etc.



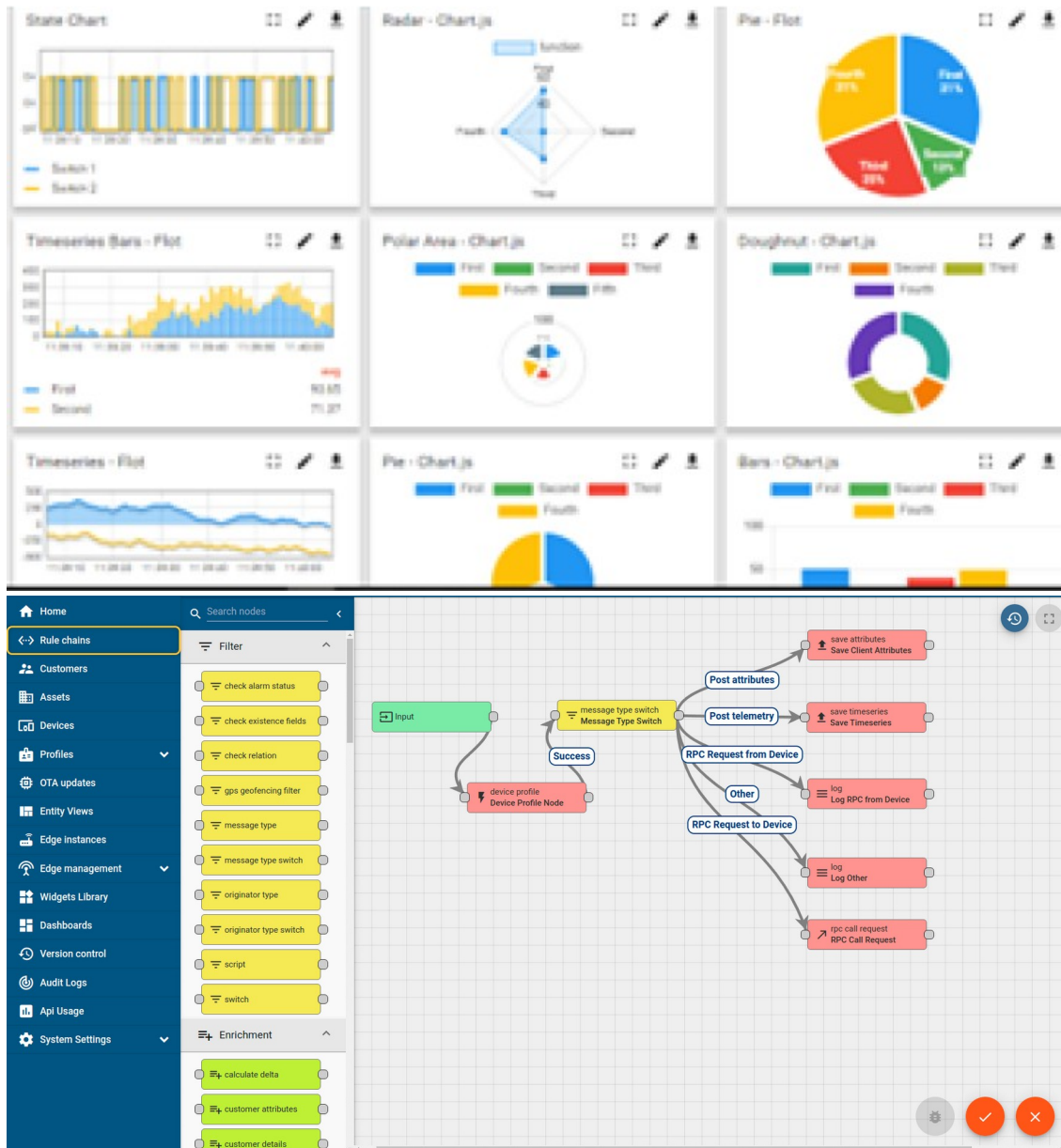
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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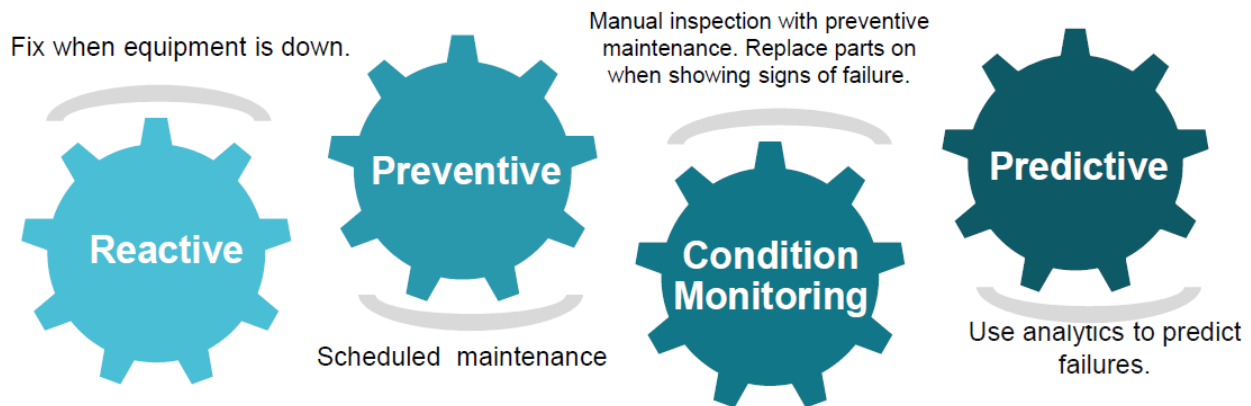


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

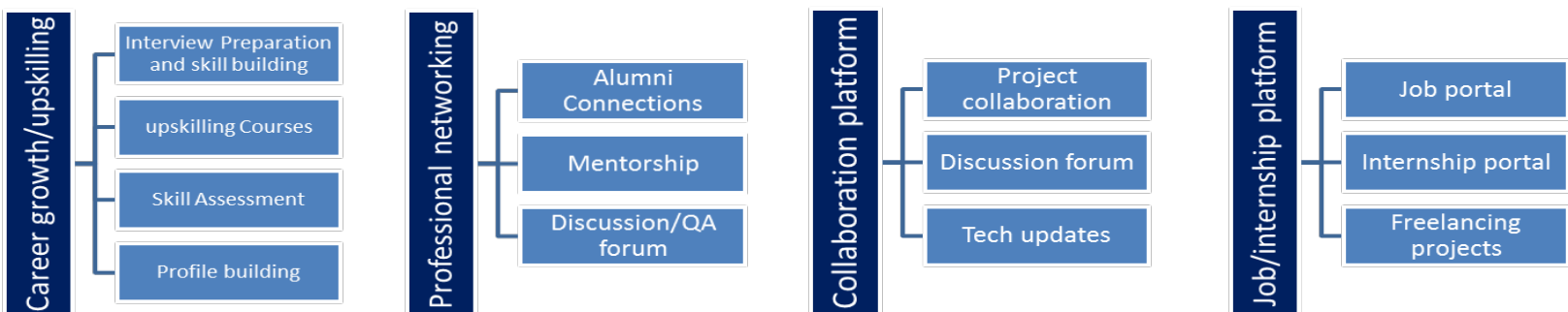
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Flask Documentation — flask.palletsprojects.com
- [2] Pandas Documentation — pandas.pydata.org/docs
- [3] Learning Python, 4th Edition — Mark Lutz

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

Students and instructors often rely on manual spreadsheet calculations to analyze academic performance, which is time-consuming and error-prone, especially as the number of records grows. There was a need for a simple, web-based tool that could take a dataset of student records and instantly generate summary statistics and performance insights, without requiring the user to write any code or use spreadsheet formulas.

4 Existing and Proposed solution

Most existing solutions rely on Excel/Google Sheets formulas and manual pivot tables, or standalone desktop analytics tools. These work for small datasets but become slow and error-prone as data grows, don't scale well for repeated use, and usually require the user to already know formulas or a separate analytics tool rather than a simple upload-and-view workflow.

I proposed a lightweight Flask web application where a user can upload a CSV of student records through a simple form; the backend uses Pandas and NumPy to compute statistics such as subject-wise averages, top performers and grade distribution, and SQLite to store submitted datasets for later reference, with the results displayed back on the page.

Compared to manual spreadsheet analysis, this solution removes the need for formulas entirely, gives instant results right after upload, keeps a record of past uploads in a database, and can later be extended with charts, custom filters and exportable reports.

4.1 Code submission (Github link)

<https://github.com/Abhay-rajyadav/upskillcampus/blob/main/StudentPerformanceAnalyticsDashboard.py>

4.2 Report submission (Github link) :

https://github.com/Abhay-rajyadav/upskillcampus/blob/main/StudentPerformanceAnalyticsDashboard_AbhayRaj_USC_UCT.pdf

4.3

5 Proposed Design/ Model

The application follows a simple three-stage flow: a user uploads a CSV file of student records through the web form (start stage); the Flask backend reads the file with Pandas and computes per-subject averages, top scorers and grade distribution using NumPy (intermediate stage); and the computed summary is rendered back to the user as a results page (final outcome).

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

Performance testing here focuses on showing this application can handle real usage, not just a single demo run — that is what separates practical, industry-relevant work from a purely academic exercise.

The main constraint identified was processing speed and correctness as the size of the uploaded CSV file increases.

This was addressed by using Pandas' vectorized operations, instead of manual Python loops, for all calculations, which keeps processing fast even for larger datasets.

On sample datasets of a few hundred to a few thousand rows, the application returned computed statistics in well under a second, with accurate averages and grade counts verified manually against a small test dataset.

Constraints considered included processing speed, memory usage for larger files, and accuracy of the computed statistics.

Testing was limited to sample datasets rather than production-scale data; for larger real-world datasets the recommendation is to add chunked file reading and background processing so the web request doesn't time out.

6.1 Test Plan/ Test Cases

Test cases covered: uploading a valid CSV with complete records, uploading a CSV with missing values, uploading a very small dataset (1-2 rows), and uploading a larger dataset (1000+ rows) to check performance and correctness of the computed statistics.

6.2 Test Procedure

Each test dataset was uploaded through the web form, and the statistics shown on the results page (averages, top performers, grade distribution) were cross-checked against the same calculations done manually in a spreadsheet for a small subset of records.

6.3 Performance Outcome

All test cases returned correct, matching statistics, and larger datasets processed noticeably faster after switching from manual loops to Pandas' built-in vectorized operations, confirming the approach scales reasonably well for the target use case.

7 My learnings

This internship taught me how to structure and build a real Flask web application end-to-end, deepened my understanding of Python fundamentals, and gave me hands-on experience using NumPy and Pandas for practical data analysis. Working through weekly assignments and a timed assessment also improved my time-management and debugging skills, all of which I expect to be directly useful in future software development roles.

8 Future work scope

Future enhancements planned for the Student Performance Analytics Dashboard include: adding chart-based visualizations (using Matplotlib or Plotly) for grade distribution, supporting custom filters (by subject or class), adding user authentication so multiple teachers can manage their own datasets, and allowing exportable PDF/Excel performance reports.